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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

HOSPITAL PATIENT REGISTRATION AND APPOINTMENT SCHEDULING SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled "HOSPITAL PATIENT REGISTRATION AND APPOINTMENT SCHEDULING SYSTEM" submitted in partial fulfillment of the requirements for the course [Course Name] is a record of my original work carried out under the guidance of [DHILEEBAN A], Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled "HOSPITAL PATIENT REGISTRATION AND APPOINTMENT SCHEDULING SYSTEM" is a bona fide work carried out by [MITHRA V] [B.Sc IT], [Information Technology], KG College of Arts & Science, in partial fulfillment of the requirements for the course [java programming].

This project has been completed under my supervision and guidance during the academic period April 2026.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

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ABSTRACT

The Hospital Patient Registration and Appointment Scheduling System is a Java-based application designed to automate patient management processes in hospitals and clinics. In many small healthcare facilities, patient registration and appointment booking are carried out manually, which often leads to scheduling conflicts, long waiting times, and poor record management.

This system enables hospital staff to register new patients, manage patient records, schedule doctor appointments, and view appointment status efficiently. The application is developed using Core Java programming concepts including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented as a console-based application or using Java Swing for a graphical user interface.

The main objective of the system is to improve patient experience, reduce appointment conflicts, and provide a structured digital platform for healthcare administration. The system generates clear appointment confirmations and organizes patient data for easy retrieval.

This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a solid foundation for developing more advanced hospital management systems in the future.

1. INTRODUCTION

In the healthcare industry, efficient patient management systems play a crucial role in delivering quality medical services and ensuring smooth hospital operations. Traditionally, many hospitals and clinics rely on manual methods for patient registration and appointment scheduling, where staff maintain handwritten registers, paper files, and simple appointment books. This manual approach is time-consuming and often leads to scheduling conflicts, misplaced records, long patient waiting times, and difficulty in tracking appointment histories.

With the rapid advancement of information technology, automated patient management systems have become essential for modern healthcare facilities. A computerized system helps hospital staff quickly register patients, assign unique patient IDs, schedule doctor appointments without conflicts, and maintain organized digital records.

The Hospital Patient Registration and Appointment Scheduling System is a Java-based application designed to automate these critical processes. The system allows hospital staff to register new patients, assign appointments with available doctors, view scheduled appointments, and cancel or reschedule as needed. The application is developed using Core Java programming concepts such as classes, objects, inheritance, methods, conditional statements, loops, and exception handling.

The primary goal of this system is to reduce manual effort, eliminate scheduling errors, improve patient experience, and provide a reliable digital platform for hospital administration.

1.1 Objective

The main objective of the Hospital Patient Registration and Appointment Scheduling System is to develop a simple and efficient patient management application using Core Java that automates the processes of patient registration, record management, and appointment scheduling in hospitals and clinics.

The specific objectives of this project are:

- To design and develop a Java-based patient registration and appointment scheduling system.
- To automate the generation of unique patient IDs for each registered patient.
- To enable hospital staff to schedule, view, and cancel doctor appointments efficiently.
- To eliminate double-booking and appointment conflicts through scheduling validation.
- To maintain organized patient records for easy retrieval and management.
- To demonstrate the implementation of Core Java programming concepts such as classes, objects, inheritance, collections, and exception handling.
- To generate clear appointment confirmations and patient registration summaries.

1.2 Overview

The Hospital Patient Registration and Appointment Scheduling System is a menu-driven application developed using the Java programming language. The system provides a digital platform where hospital staff can manage all patient-related operations from a single interface.

The system allows staff to register new patients by entering basic information such as name, age, gender, contact details, and medical history. Each patient is automatically assigned a unique Patient ID for future reference. Staff can then schedule appointments by selecting the patient, choosing a doctor, and selecting an available date and time slot.

The system validates appointment requests to avoid scheduling conflicts and confirms successful bookings. Staff can also view all scheduled appointments, search for patient records, and cancel or reschedule appointments as required.

This project demonstrates the practical use of Java programming in developing a real-world healthcare administration application suitable for small hospitals, clinics, and healthcare centers.

1.3 Features

The Hospital Patient Registration and Appointment Scheduling System includes the following main features:

- Patient Registration – Registers new patients with complete personal and medical details, and assigns a unique Patient ID automatically.
- Appointment Scheduling – Allows staff to schedule doctor appointments by selecting patient, doctor, date, and time slot.
- Conflict Detection – Validates appointment requests to prevent double-booking of doctors and time slots.
- Appointment View – Displays all scheduled appointments with patient name, doctor, date, and time details.
- Patient Record Management – Allows staff to search, view, and update patient records by Patient ID.
- Appointment Cancellation – Enables cancellation and rescheduling of existing appointments.
- User-Friendly Interface – Provides a simple menu-driven interface for hospital staff.
- Unique ID Generation – Automatically generates unique Patient IDs for each registered patient.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in existing manual hospital administration systems and proposing a computerized solution that improves efficiency and accuracy in patient management.

2.1 Problem Definition

In many small hospitals and clinics, patient registration and appointment scheduling are often performed manually using paper registers and appointment books. This traditional method creates several challenges.

Some common problems include:

- Double-booking of doctors due to lack of automated conflict detection.
- Long patient waiting times due to poor appointment management.
- Misplaced or lost patient records in paper-based filing systems.
- Difficulty in quickly retrieving patient history during consultations.
- Lack of a centralized system to manage multiple doctors and departments.
- Inefficient use of staff time due to manual administrative tasks.

Manual patient management systems also increase the workload on hospital administrative staff and can lead to patient dissatisfaction due to scheduling errors, delays, and poor record keeping.

Therefore, there is a clear need for an automated patient registration and appointment scheduling system that can efficiently manage patient data, schedule appointments without conflicts, and provide quick access to patient records.

2.2 Project Overview

The Hospital Patient Registration and Appointment Scheduling System is a Java-based application developed to streamline patient management in hospitals and clinics.

The system provides a menu-driven interface where hospital staff can perform the following functions:

- Register new patients and generate unique Patient IDs.
- Schedule appointments with available doctors for selected date and time slots.
- View all scheduled appointments in an organized format.
- Search and retrieve patient records using Patient ID.
- Cancel or reschedule existing appointments.

This project is implemented using Core Java programming concepts and can be developed as either a console-based program or a Java Swing GUI application.

2.3 Scopes and Limitations

Scope

The scope of the Hospital Patient Registration and Appointment Scheduling System includes:

- Managing complete patient registration records.
- Processing and validating doctor appointment requests.
- Preventing appointment conflicts through scheduling validation.
- Maintaining patient medical history summaries.
- Generating appointment confirmations and patient summaries.
- Demonstrating the practical use of Java OOP concepts.

The system can be used in:

- Government and private hospitals.
- Clinics and health centers.
- Diagnostic and specialty care centers.
- Dental and eye care clinics.

Limitations

- The system does not include a database; all records are stored in memory and are lost after the program exits.
- Online or remote appointment booking is not supported in the basic version.
- Payment and billing integration is not included.
- The system does not generate printed appointment cards or SMS notifications.

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes the development process easier.

The Hospital Patient Registration and Appointment Scheduling System is designed as a menu-driven application where hospital staff can register patients, schedule appointments, and manage all related records from a single interface.

The system is divided into several modules, each responsible for handling a specific function.

3.1 Modules

The main modules of the Hospital Patient Registration and Appointment Scheduling System are explained below:

1. Patient Registration Module

This module handles the registration of new patients. When a new patient visits the hospital, staff enter the following details:

- Patient Name
- Age and Gender
- Contact Number
- Address
- Known Medical Conditions / Chief Complaint

The system automatically generates a unique Patient ID (e.g., PID001, PID002) for each registered patient. Patient data is stored in a list for retrieval during appointment scheduling.

2. Doctor Management Module

This module manages the list of available doctors. Each doctor record includes:

- Doctor Name
- Specialization (e.g., General Medicine, Cardiology, Orthopedics)
- Doctor ID
- Available consultation days and time slots

The module ensures that each doctor's available slots are tracked and updated after every appointment booking.

3. Appointment Scheduling Module

This is the core module of the system. It allows hospital staff to schedule doctor appointments for registered patients.

Functions of this module include:

- Selecting a registered patient by Patient ID.
- Selecting a doctor from the available list.
- Choosing an appointment date and available time slot.
- Validating the slot to prevent double-booking.
- Generating a unique Appointment ID (e.g., APT001).
- Confirming and saving the appointment details.

4. Appointment View Module

This module displays all scheduled appointments in an organized format. Each appointment record includes:

- Appointment ID
- Patient Name and Patient ID
- Doctor Name and Specialization
- Appointment Date and Time
- Status (Scheduled / Cancelled)

5. Appointment Cancellation Module

This module allows staff to cancel an existing appointment by entering the Appointment ID. Upon cancellation, the time slot is freed for new bookings.

6. Patient Search Module

This module allows hospital staff to search for patient records using the Patient ID. It displays the patient's registration details and appointment history.

7. Exit Module

This module allows the user to exit the application safely after completing all required operations.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Hospital Patient Registration and Appointment Scheduling System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages: Problem Analysis, System Design, Program Development, Testing and Validation, and Result and Documentation.

4.1 Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the operational needs of hospital administration.

The main problems identified in manual patient management systems are:

- Manual registration processes are time-consuming and error-prone.
- Appointment conflicts arise due to lack of automated scheduling validation.
- Paper-based patient records are difficult to retrieve quickly during consultations.
- Multiple staff members cannot efficiently coordinate patient scheduling manually.

Based on these problems, the system requirements were defined to develop an automated patient management application that simplifies hospital administrative operations.

4.2 System Design

In this stage, the overall structure of the system is designed before coding begins. The system is divided into modules such as: Patient Registration Module, Doctor Management Module, Appointment Scheduling Module, Appointment View Module, Appointment Cancellation Module, Patient Search Module, and Exit Module.

The program flow and data structures are designed so that hospital staff can easily interact with the system through a clear menu-driven interface.

4.3 Program Development

After designing the system, the program is implemented using the Java programming language. The following steps were followed during development:

- Creation of Patient and Doctor classes with appropriate attributes and methods.
- Implementation of the AppointmentScheduler class to handle booking logic.
- Use of ArrayList to store patient and appointment records.
- Implementation of menu-driven logic using loops and switch statements.
- Writing code for Patient ID and Appointment ID generation.
- Implementing slot validation to prevent double-booking.
- Handling invalid user inputs using try-catch exception handling.

Core Java concepts such as classes, objects, inheritance, ArrayList, methods, loops, conditional statements, and exception handling were applied during development.

4.4 Testing and Validation

Testing ensures that the program works correctly and produces the expected results. Different test cases were used to check the functionality of the system.

Test Case	Expected Result
Register a new patient	Patient registered with unique Patient ID
Schedule appointment for valid patient	Appointment confirmed with unique Appointment ID
Book same doctor, same slot twice	Conflict detected, booking rejected
View all appointments	All appointments displayed correctly
Cancel an appointment by ID	Appointment cancelled, slot freed
Search patient by invalid ID	Error message displayed
Enter non-numeric input for age	Exception handled, prompt re-displayed

This phase ensures that the program runs without errors and produces accurate, reliable results under all test conditions.

4.5 Result and Documentation

After successful testing, the results of the system are documented. The system successfully registers patients, schedules appointments with conflict detection, and displays all appointment records accurately.

The results are demonstrated through program execution and screenshots showing the input and output of the system. Proper documentation is prepared to explain the design, implementation, and working of each module.

4.6 Algorithm

Algorithm for the Hospital Patient Registration and Appointment Scheduling System:

1. Start the program.
2. Display the main menu with options: Register Patient, Schedule Appointment, View Appointments, Search Patient, Cancel Appointment, Exit.
3. If option 1 selected: Accept patient details, generate unique Patient ID, save patient record.
4. If option 2 selected: Accept Patient ID, validate patient exists, display available doctors, accept doctor selection, accept desired date and time slot, check for scheduling conflict.
5. If slot is available: Confirm appointment, generate Appointment ID, save appointment record.
6. If slot is not available: Display conflict message, ask user to choose a different slot.
7. If option 3 selected: Retrieve and display all scheduled appointments.
8. If option 4 selected: Accept Patient ID, retrieve and display patient record and appointment history.
9. If option 5 selected: Accept Appointment ID, mark appointment as cancelled, free the slot.
10. If option 6 selected: Exit the application.

11. Repeat the menu until exit is selected.
12. End the program.

4.7 Flowchart

Students should draw a flowchart showing the program flow for the Hospital Patient Registration and Appointment Scheduling System.

Example flow:

Start → Display Main Menu → User Selects Option
↓ Option 1: Register Patient → Input Details → Generate ID → Save Record
↓ Option 2: Schedule Appointment → Select Patient → Select Doctor
→ Select Date/Time → Check Conflict → Confirm or Reject
↓ Option 3: View Appointments → Display All Records
↓ Option 6: Exit → End

Students can draw the flowchart using draw.io, Lucidchart, or MS Visio.

5. SOURCE CODE

Main class

```
package com.Structure;

import java.util.Scanner;

public class Main {
   public static void main(String[] args) {

      Scanner sc = new Scanner(System.in);
      HospitalService service = new HospitalService();

      while (true) {
         System.out.println("\n1. Add Patient");
         System.out.println("2. View Patients");
         System.out.println("3. Book Appointment");
         System.out.println("4. View Appointments");
         System.out.println("5. Exit");

         int choice = sc.nextInt();

         switch (choice) {

            case 1:
               System.out.print("Enter ID: ");
               int id = sc.nextInt();
               sc.nextLine();

               System.out.print("Enter Name: ");
               String name = sc.nextLine();

               System.out.print("Enter Age: ");
               int age = sc.nextInt();
```

```
                service.addPatient(new Patient(id, name, age));
                break;

            case 2:
                service.viewPatients();
                break;

            case 3:
                System.out.print("Enter Appointment ID: ");
                int aid = sc.nextInt();
                sc.nextLine();

                System.out.print("Enter Patient Name: ");
                String pname = sc.nextLine();

                System.out.print("Enter Doctor Name: ");
                String dname = sc.nextLine();

                System.out.print("Enter Date: ");
                String date = sc.nextLine();

                service.bookAppointment(new Appointment(aid, pname, dname, date));
                break;

            case 4:
                service.viewAppointments();
                break;

            case 5:
                System.out.println("Thank You!");
                return;

            default:
                System.out.println("Invalid Choice");
        }
    }
  }
}
```

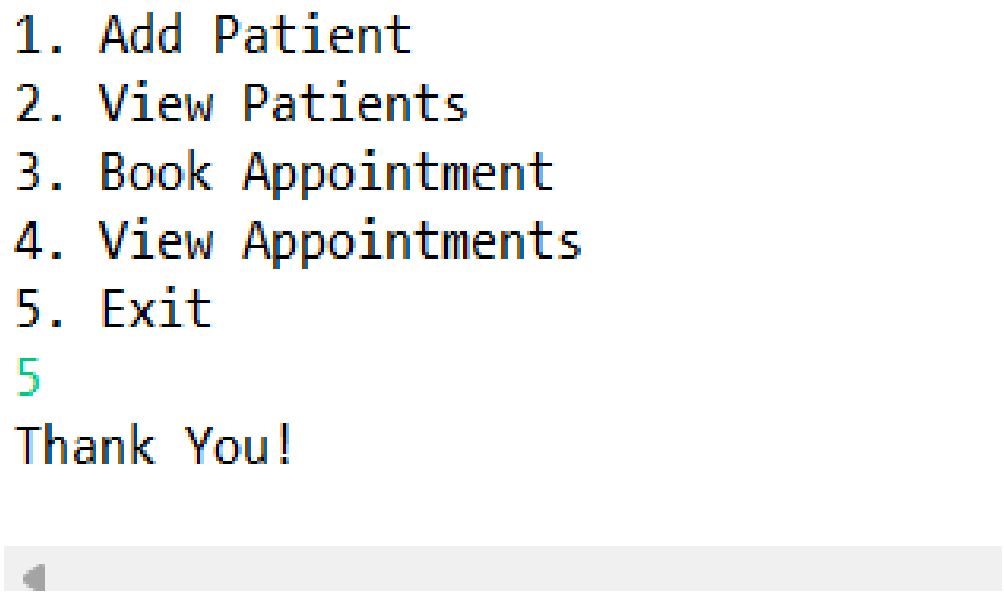
6. INPUT AND OUTPUT SCREENS

```
1. Add Patient
2. View Patients
3. Book Appointment
4. View Appointments
5. Exit
1
Enter ID: 102
Enter Name: mithra
Enter Age: 18
Patient Registered Successfully

1. Add Patient
2. View Patients
3. Book Appointment
4. View Appointments
5. Exit
2
102 mithra 18


 1. Add Patient
 2. View Patients
 3. Book Appointment
 4. View Appointments
 5. Exit
 3
 Enter Appointment ID: 102
 Enter Patient Name: mithra
 Enter Doctor Name: gopika
 Enter Date: 20-08-2026
 Appointment Booked Successfully

 1. Add Patient
 2. View Patients
 3. Book Appointment
 4. View Appointments
 5. Exit
 4
 102 mithra gopika 20-08-2026
```



7. DISCUSSION AND CONCLUSION

7.1 Discussion

The Hospital Patient Registration and Appointment Scheduling System was developed to automate the administrative processes of patient management in hospitals and clinics. The system successfully enables hospital staff to register patients, schedule appointments, detect scheduling conflicts, and manage appointment records efficiently.

During the development process, Core Java programming concepts such as classes, objects, inheritance, ArrayList, loops, conditional statements, methods, and exception handling were applied. The use of object-oriented design made the system modular and easy to maintain, with separate classes for Patient, Doctor, and Appointment entities.

The scheduling conflict detection feature is a key strength of this system. By validating each appointment request against existing bookings for the same doctor, date, and time slot, the system ensures that no double-booking occurs, which is a critical requirement in a real hospital environment.

The system was tested with multiple test cases including normal registrations, valid and invalid appointment requests, and conflict scenarios. The results confirmed that the system handles all cases correctly and produces accurate outputs.

The major limitation of the current version is that all data is stored in memory and is not persisted to a file or database. This means records are lost when the program is closed. Future versions could address this by integrating a database such as MySQL using JDBC.

7.2 Conclusion

The Hospital Patient Registration and Appointment Scheduling System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world healthcare administration system.

The system automates patient registration, generates unique patient identifiers, manages appointment scheduling with conflict detection, and provides efficient access to patient and appointment records. It significantly reduces manual effort and eliminates common administrative errors found in paper-based hospital management systems.

This project helps students understand how Java programming can be applied to solve real-world business and administrative problems using object-oriented design principles. The project can be further enhanced by adding features such as database connectivity using JDBC, a graphical user interface using Java Swing or JavaFX, SMS and email appointment reminders, and an online patient portal for self-booking.

8. REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – Java: The Complete Reference (12th Edition), McGraw-Hill.
- Cay S. Horstmann – Core Java Volume I – Fundamentals, Pearson Education.
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Websites

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- https://docs.oracle.com/javase – Java SE API Documentation

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